CIND 820: Big Data Analytics Project

Analyzing Clickstream Data from an E-commerce Website through Market Basket, Cluster, and Classification Analysis Techniques

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Dataset: Clickstream Data for Online Shopping



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Abstract

Digital marketing data can provide valuable intelligence and competitive advantages in today's highly competitive digital business landscape. One of the key benefits of digital marketing is that it is easy to track and measure (*What Are the 8 Types of Digital Marketing?*, 2022). With this comes big data, where millions of page views, sessions, and user data points are collected daily from sites that can be strategically leveraged to achieve data-driven sales and create more significant business gains. In the wide breadth of digital marketing data, clickstream data is an important and informative subsection that provides insight into how consumers navigate a site, what pages consumers are most interested in, and what pages are the most essential in consumers reaching desired target pages (Albert et al., 2010). Optimization and improved user experience and interfaces can differentiate between a successful conversion and a missed opportunity. In this analysis, clickstream data will be analyzed using unsupervised and supervised machine learning techniques such as classification and clustering and the data mining technique, market basket analysis.

In this report, the dataset titled “Clickstream Data for Online Shopping” will be researched (Lapczynski & Białowąs, 2013). This 14-variable dataset contains clickstream information from an online store that sells maternity clothing for pregnant women. The data was collected for five months in 2008 (Lapczynski & Białowąs, 2013).

Research areas of interest include determining which product items are highly associated with each other as they are viewed together during customer sessions on the e-commerce store. This intelligence can spark all kinds of strategic implementations, like bundling offers that can increase customer order values. Market basket analysis can be investigated using association rules and the Apriori algorithm, Eclat Algorithm and Weclat Algorithm. These algorithms use

for mining frequent item sets and relevant association rules. It uses prior knowledge of recurring itemset properties and employs an iterative approach (Li, 2009). The basket analysis will be performed using the Apriori algorithm, Eclat Algorithm and Weclat Algorithm in.

The second area of interest is clustering to determine customer segments and discover characteristics amongst similar customers. This technique can be used to understand customer behaviour better and cater to customers in a more personalized way. Cluster analysis, an unsupervised machine learning technique, allows organizations to better understand their customers by identifying individuals with similar traits, which can inform how the organization communicates with those customers (*Cluster Analysis*, 2021). Cluster analysis provides insight into the data by dividing the objects into groups (clusters) of objects such that objects in a cluster are more similar than objects in other clusters (Wu, 2012). The cluster analysis for this dataset will be performed using the K-means clustering and Kmode clustering in python and r.

The last area of interest is using predictive analytics via classification to predict product prices based on past/training data. Predicted prices can be used to price items strategically based on previous product listings, such as pricing items more highly if past data shows similar products were able to be upsold based on additional perceived value by the customer for certain features of the products. Classification analysis, a supervised machine learning technique, determines the class label for an unlabeled test data point given a set of training data points along with associated training labels (Aggarwal, 2014). The classification analysis for this dataset will be performed using python to create classification decision trees. The data will be split using 10-fold cross validation to train the model and create a classification decision trees to predict whether an item will be priced higher than the average price for the entire product category. This will be accomplished using rpart in r and HPsplit in SAS.

This report will employ machine learning and data mining techniques to analyze clickstream data and gain more insight into product pricing, associated products, and customer segmentation through classification, basket analysis, and clustering techniques.

Literature Review

Data mining is an emerging field with multidisciplinary research. It combines disciplines such as databases, machine learning, artificial intelligence, statistics, automated scientific discovery, data visualization, decision science, and high-performance computing. Data mining techniques can help users find relevant and applicable information on the internet, automate the analysis of users' navigation, and be combined to create data-mined knowledge of multiple users' purchasing behaviour (Zhang et al., 2007). While a customer is visiting a site, they leave a trail of their information via a web log. This web log is a valuable resource to gain more information about customer behaviour. This has potential to help aid strategic planning activities when targeting particular customer groups for new products based on user behaviour (Zhang et al., 2007).

Web mining is the exploration and analysis of information from the world wide web, that can help provide key insight into user behavior and preferences from generating enormous amounts of datapoints. This works by searching for information through online resources also known as web content mining. Similarly, web usage mining unearths users' online behaviour, activities and patterns from web logs (Lapczynski & Białowąs, 2013). This can provide a great deal of insight to marketing researchers and key stakeholders invested in customer relationships

so they can better understand consumer purchasing behaviour and make the appropriate business decisions accordingly. Web mining plays a crucial role in discovering and analyzing clickstream datasets. The "Clickstream Data for Online Shopping" dataset was created by collecting and merging web clickstream data from web mining with other features and attributes collected from users while visiting other websites on the internet (Lapczynski & Białowąs, 2013). Data mining has countless applications such as understanding consumer behaviour in digital spaces, determining price elasticity of products and goods as well as stock management in retail settings, and promotional activities like digital campaigns (Lapczynski & Białowąs, 2013).

The "Clickstream Data for Online Shopping" dataset used in this big data analytics project originates from a published research paper titled "Discovering Patterns of Users' Behaviour in an E-shop – Comparison of Consumer Buying Behaviours in Poland and Other European Countries" (Lapczynski & Białowąs, 2013). In this research paper, the ecommerce store's log files were used to extract the dataset; it consists of 210,814 rows. The dataset referenced the different clothing categories like skirts, dresses, trousers, blouses, and special offers. Since, the website had translations for multiple languages including English, German and Czech and can provide shipping abroad, customers could derive from other European countries outside Poland (Lapczynski & Białowąs, 2013). To gather country location, website visitor IP addresses were obtained from web logs, and then transformed using geolocation to return the country of the visitor. In some cases, this information wasn't available as the geolocation system used sometimes specified the continent of Europe instead of a specific country. Thus, the domain name suffix was used to group by if it ended with ".net," ".com," ".biz," or ".org" (Lapczynski & Białowąs, 2013).

In Lapczynski and Białowąs's research paper, descriptive statistics, association rules and market basket analysis were used to note differences in consumer behaviour and consumer preferences by region i.e. Poland, European Union, Non-European Union, and Non-European visitors, and by product types such as blouses and tunics (Lapczynski & Białowąs, 2013).

Poland		UE		Non-EU countries		Countries outside Europe	
Type of blouse	Percent	Type of blouse	Percent	Type of blouse	Percent	Type of blouse	Percent
C5 	4.8	C5 	4.3	C56 	4.9	C35 	9.3
C17 	4.0	C17 	3.7	C6 	4.9	C38 	7.4
C14 	3.2	C56 	3.6	C2 	3.9	C2 	7.4

Figure 1: The percentage of users choosing different models of blouses, obtained from "Discovering Patterns of Users' Behaviour in an E-shop – Comparison of Consumer Buying Behaviours in Poland and Other European Countries" (Lapczynski & Białowąs, 2013).

In the figure above, the descriptive statistics indicate that the most selected clothing designs among Poland users are C5, C17 and C14, and the most selected colours are white, grey and colourful. Clothing models C5, C17 and C56 were the most popular styles chosen by European Union visitors, clothing models C56, C6 and C2 were selected most by the Non-

European Union countries, and models C35, C38, and C2 were most popular with users outside of Europe (Lapczynski & Białowąs, 2013).

Association rules are a type of unsupervised learning. The Apriori algorithm is an algorithm used for association rule learning and first came into popularity in 1993 (Lapczynski & Białowąs, 2013). Association rules are used for market basket analysis and help discover relationships between items in frequent itemsets (Supriyadi et al., 2018).

Rule 1			Rule 2		
LHS	==>	RHS	LHS	==>	RHS
C12 	==>	C17 	C17 	==>	C5 
Support = 2.97%, confidence = 37.72%, lift = 2.9			Support = 2.84%, confidence = 21.93%, lift = 1.5		

Figure 2: Association rules with the highest value of support for blouses chosen by users from Poland, obtained from "Discovering Patterns of Users' Behaviour in an E-shop – Comparison of Consumer Buying Behaviours in Poland and Other European Countries" (Lapczynski & Białowąs, 2013).

As per the figure above, the research paper examined association rules and determined commonly selected clothing models together by region and product type of blouses and tunic. The most common itemset in blouses category is item C12 and C17, followed by C17 and C5 with Poland Users (Lapczynski & Białowąs, 2013).

Another example of an application utilizing the Apriori algorithm can be found in the research paper "Website Content Analysis Using Clickstream Data and Apriori Algorithm" (Supriyadi et al., 2018). In this paper, traditional association rule mining was used via the Apriori

algorthim to discover the frequent itemsets, which defines the itemsets of each transaction in the transactional database of clickstream data and web content data (Supriyadi et al., 2018).

The analysis in this project differs by looking at the data at a higher level and holistic approach to determine customer segments through clustering, as well as price prediction for pricing new products with similar characteristics and features using classification with diving the entire dataset into training and test models. Also, analyzing what products were selected together the most frequently via market basket analysis without isolating the analysis by product type. In addition, basket analysis can also be used to see what product clothing models were selected by product type for the other product type categories such as trousers, skirts and sale items.

Clustering is a data mining technique used to analyse a large quality of data with many variations. Clustering is the process of grouping data into clusters, so the data within a cluster is as similar as possible and different from other cluster groups (Syakur et al., 2018). An example of a cluster analysis approach to understanding customer behaviours and creating consumer segments can be found in the research paper titled “Customer Segment Prediction on Retail Transactional Data Using K-Means and Markov Model” (S. Harish & Malathy, 2023). In this paper, historic retail transactional data was analyzed using data mining and cluster analysis techniques to create customer segments using K-means clustering. As a different type of dataset compared to clickstream data, transactional retail data both allow companies to segment customers usings different forms of data. However, this paper goes into more depth by predicting the cluster movements over the time period using the Markov Model algorithm (S. Harish & Malathy, 2023).

Decision trees are a type of supervised machine learning technique; it is used for building classification models. Decision trees are tree-like structures where there are leaf nodes that provide the outcome, and internal nodes that differ from the leaf nodes that are decision nodes. For each of the decision nodes, it can create a further split based on each classification on a no/yes question; such data splits can be determined using different methods such as entropy. This classification model helps predict the target variable by training it on simple decision rules based on a given dataset (Al-Turjman et al., 2022). In this research paper "Vehicle Price Classification and Prediction Using Machine Learning in the IoT Smart Manufacturing Era" supervised machine learning strategies such as classification was used to predict vehicle prices using historic labelled data. Similar to this big data analytics project's approach, decision trees were used as a method of classification (Al-Turjman et al., 2022).

Research Questions

Market Basket Analysis

1. Which products are viewed together in a session?
2. What is the best product to propose to the customer?

Classification

3. What will future pricing predictions look like based on current data? Whether it's below or above the average cost of a clothing model?

Classification

4. What customer segments can be discovered based on similar customer session behaviour?

Data Description

Table 1: A summary of all attributes in the clickstream dataset including data types and descriptive statistics.

Attributes	Short Description	Data Type	Min.	Max.	Mean	STD
1. year	from 2008	Quantitative; Discrete	2008	2008	2008	0.0
2. month	from April (4) to August (8)	Qualitative; Nominal	4	8	5.59	1.33
3. day	day number of the month	Qualitative; Nominal	1	31	14.52	8.83
4. order	sequence of clicks during one session	Qualitative; Ordinal	1	195	9.82	13.48
5. country	variable indicating the country of origin of the IP address; 47 categories (see appendix for more information)	Qualitative; Nominal	1	47	26.95	7.15
6. session ID	variable indicating session id (short record)	Quantitative; Discrete	1	24026	12058.41	7008.42
7. page 1 (main category)	concerns the main product category: 1-trousers, 2-skirts, 3-blouses, 4-sale	Qualitative; Nominal	1	4	2.40	1.14
8. page 2 (clothing model)	contains information about the code for each product i.e. A14, B4, C56, P1	Qualitative; Nominal	N/A	N/A	N/A	N/A
9. colour	colour of product; 14 categories (see appendix for more information)	Qualitative; Nominal	1	14	6.23	4.24
10. location	photo location on the page, the screen has been divided into six parts: 1-top left, 2-top in the middle, 3-top right, 4-bottom left, 5-bottom in the middle, 6-bottom right	Qualitative; Nominal	1	6	3.26	1.71
11. model photography	variable with two categories: 1-en face, 2-profile	Qualitative; Nominal	1.0	2.0	1.26	0.44
12. price	price in US dollars	Quantitative; Continuous	18.0	82.0	43.80	12.55
13. price 2	variable informing whether the price of a particular product is higher than the average price for the entire product category: 1-yes, 2-no	Qualitative; Nominal	1.0	2.0	1.49	0.50
14. page	page number within the e-store website (from 1 to 5)	Qualitative; Ordinal	1.0	5.0	1.71	0.98

Since the features of the dataset include those that are qualitative, their respective mean and standard deviation is not meaningful in this application and such values are denoted in the table above in

grey font for the standard deviation and mean for qualitative values as mean and standard deviation is not an appropriate statistic to be applied to qualitative data.

Approach



Figure 3: A flowchart representing the approach for this big data analytics project.

Data Cleaning & Preparation

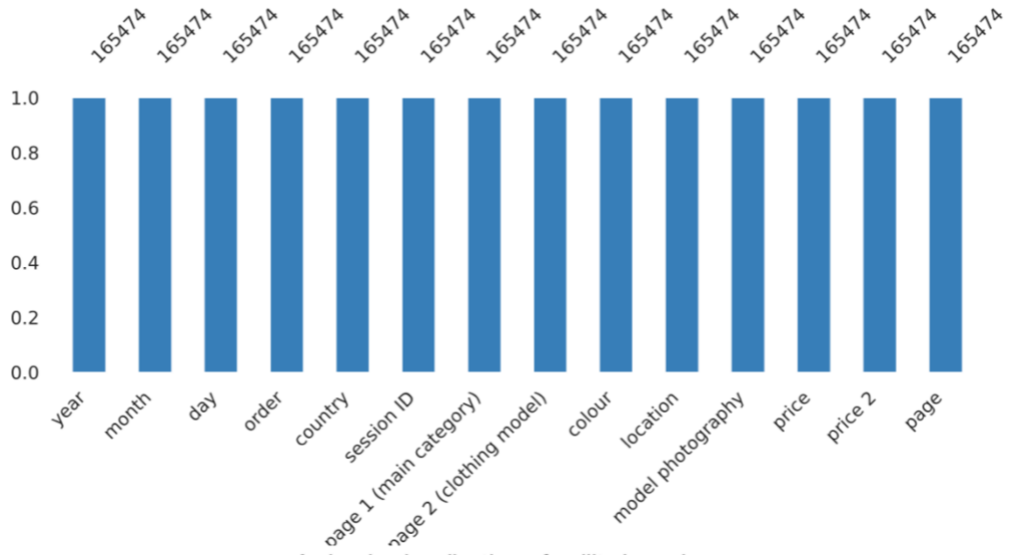


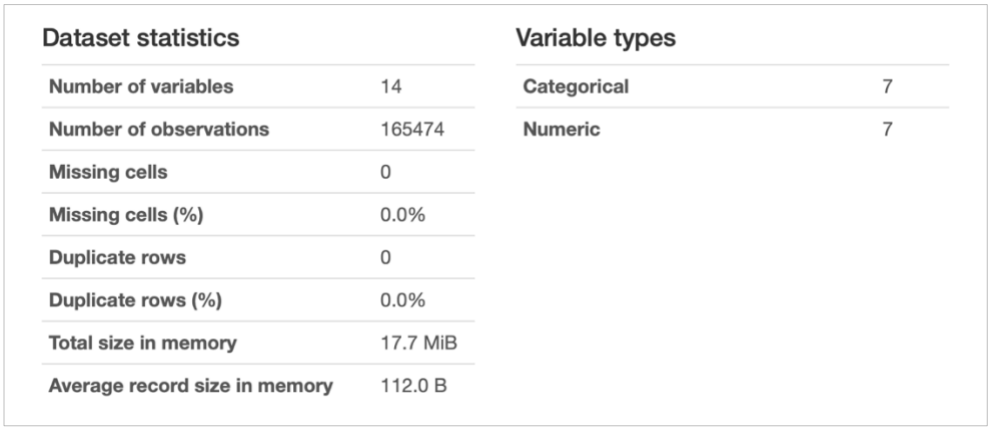
Figure 4 A simple visualization of nullity by feature.

The dataset was found to not contain any NULLs, NAs, or missing values. The dataset was also found to not have any duplicates.

Exploratory Data Analysis

Exploratory analysis is generally presented for each attribute. Charts were prepared via exploratory data analysis report in python.

Summary



Dataset statistics	
Number of variables	14
Number of observations	165474
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	17.7 MiB
Average record size in memory	112.0 B

Variable types	
Categorical	7
Numeric	7

Figure 5: Summary of dataset statistics

Based on the correlation plot below, the following pairs of attributes appear to be correlated:

- Month & Session ID
- Price & Price 2
- Page 1 (main category) & Colour
- Model Photography & Colour
- Price 2 & Colour
- Page & Colour

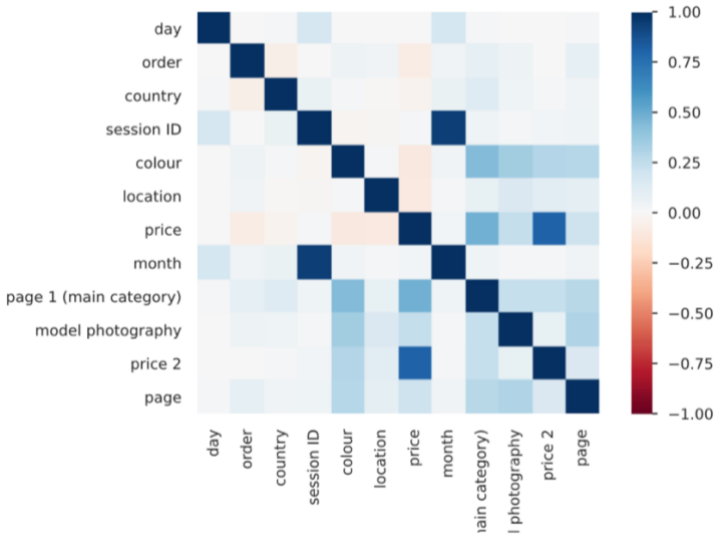


Figure 6: Correlation plot of 14 attributes in the clickstream

1. Year

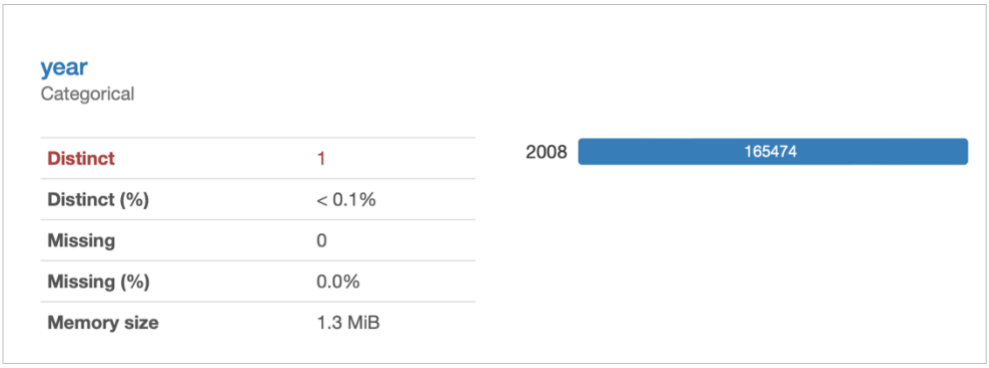


Figure 7: Summary of the Year variable.

2. Month

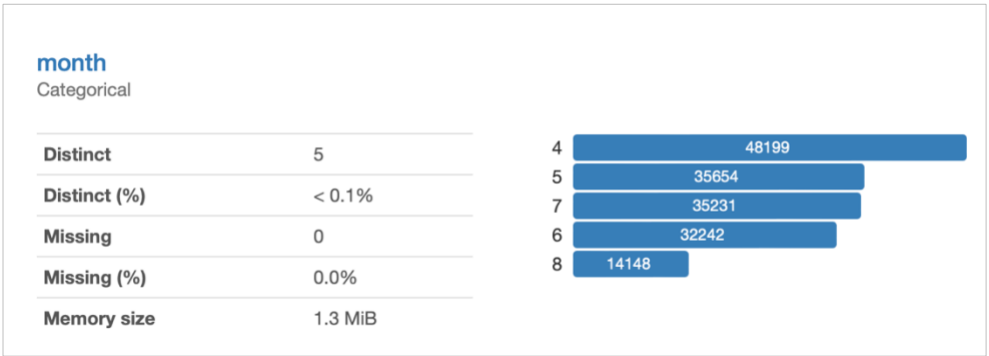


Figure 8: Summary of the Month variable.

3. Day

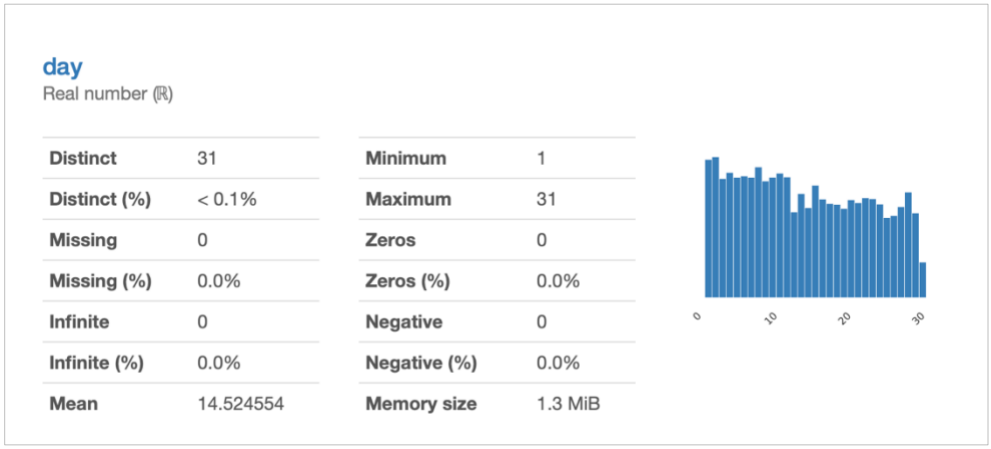


Figure 9: Summary of the Day variable.

4. Order

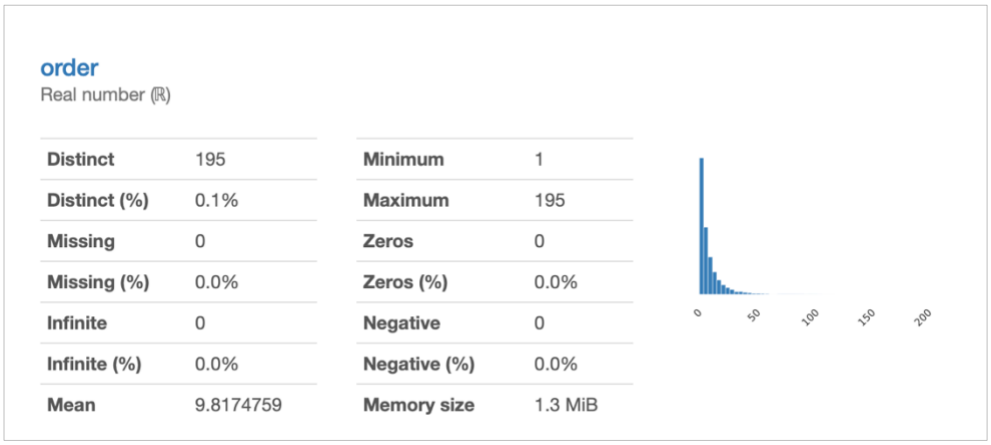


Figure 10: Summary of the Order variable.

5. Country

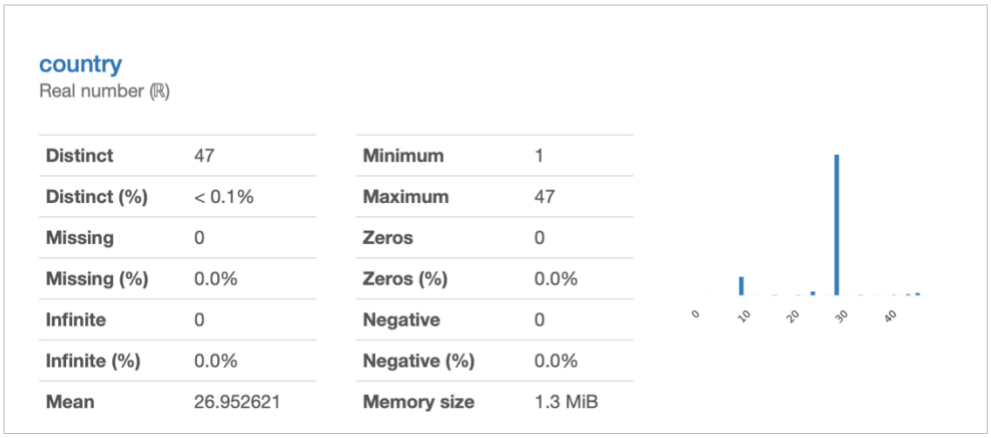


Figure 11: Summary of the Country variable.

6. Session ID

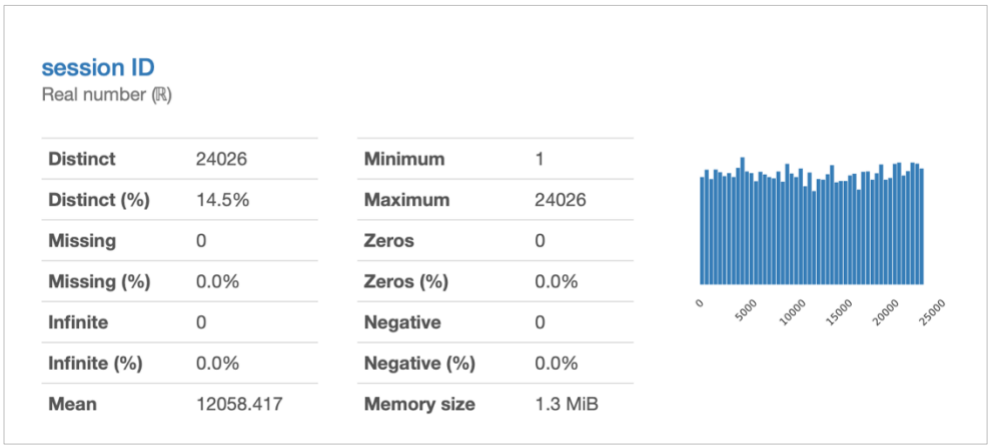


Figure 12: Summary of the Session ID variable.

7. Page 1 (main category)

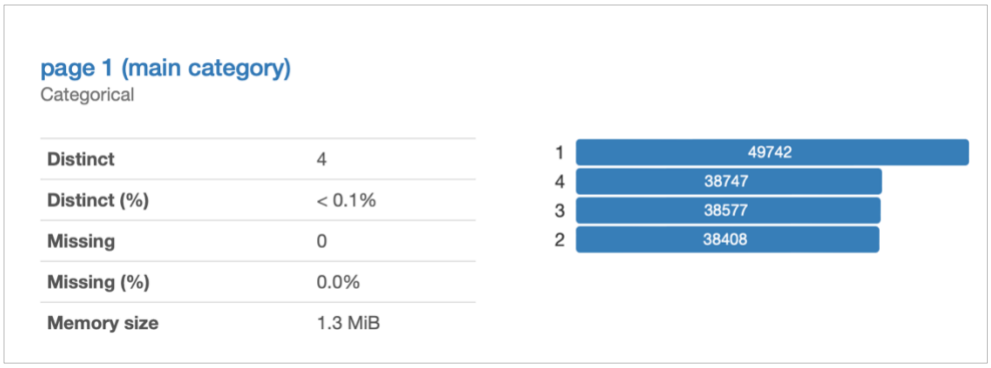


Figure 13: Summary of the Page 1 (main category) variable.

8. Page 2 (Clothing Model)

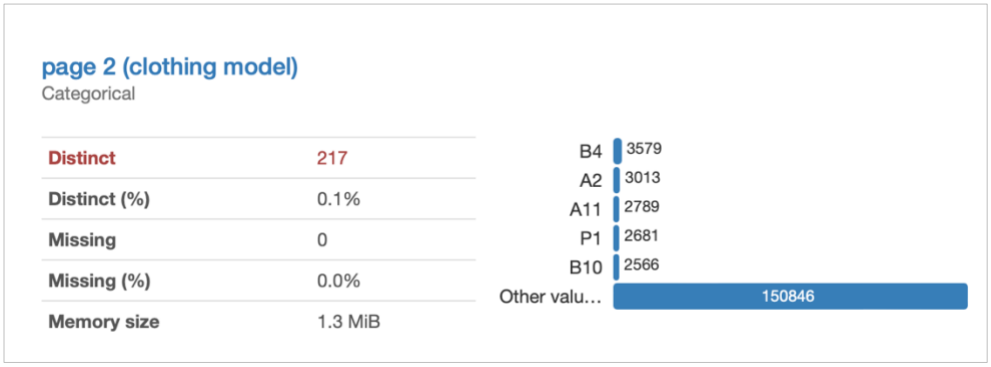


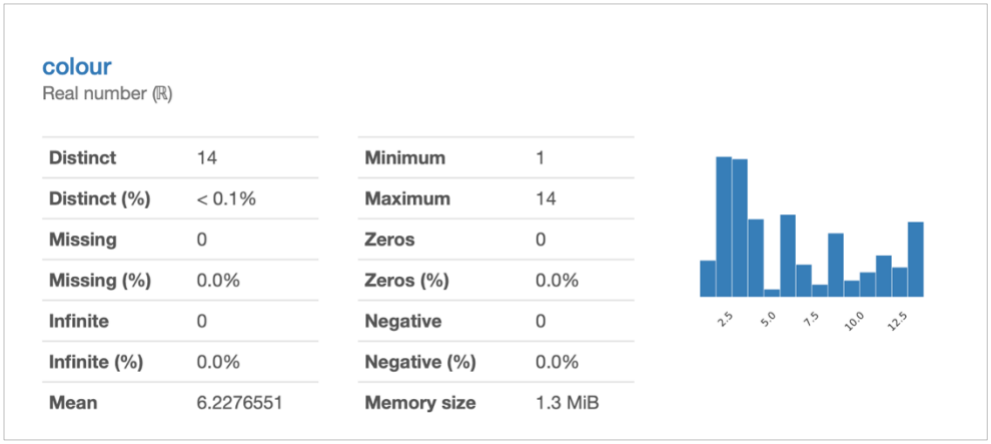
Figure 14: Summary of the Page 2 (Clothing Model) variable.

Common Values

Value	Count	Frequency (%)
B4	3579	2.2%
A2	3013	1.8%
A11	2789	1.7%
P1	2681	1.6%
B10	2566	1.6%
A4	2522	1.5%
A15	2489	1.5%
A5	2354	1.4%
A10	2280	1.4%
A1	2265	1.4%
Other values (207)	138936	84.0%

Figure 15: Top 10 Frequencies of Page 2 (Clothing Models)

9. Colour

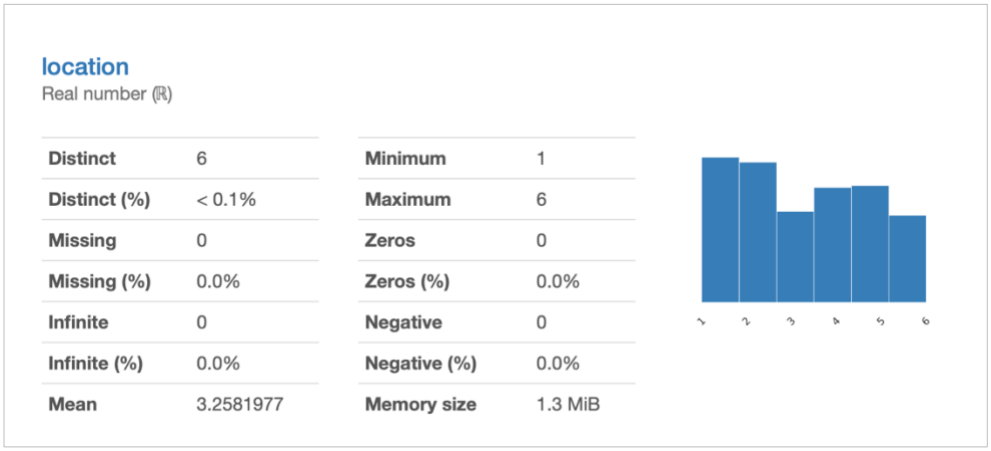


colour
Real number (ℝ)

Distinct	14	Minimum	1
Distinct (%)	< 0.1%	Maximum	14
Missing	0	Zeros	0
Missing (%)	0.0%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	6.2276551	Memory size	1.3 MiB

Figure 16: Summary of the Colour variable.

10. Location



location
Real number (ℝ)

Distinct	6	Minimum	1
Distinct (%)	< 0.1%	Maximum	6
Missing	0	Zeros	0
Missing (%)	0.0%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	3.2581977	Memory size	1.3 MiB

Figure 17: Summary of the Location variable.

11. Model Photography

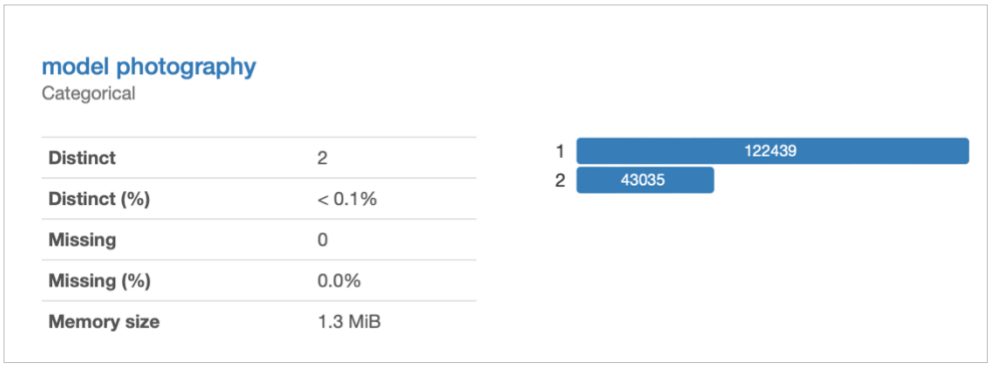


Figure 18: Summary of the Model Photography variable.

12. Price

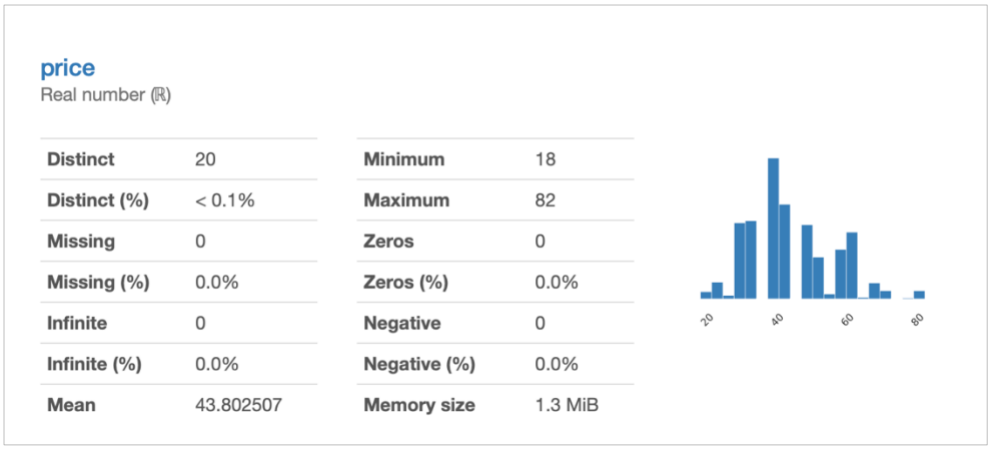


Figure 19: Summary of the Price variable.

13. Price 2

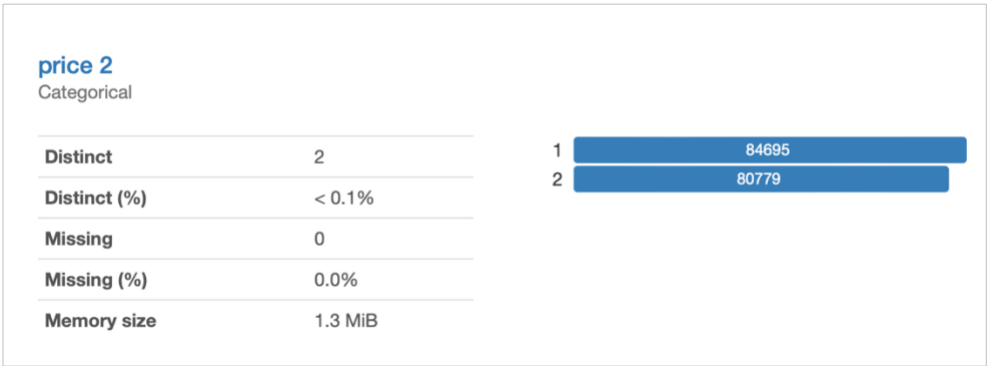


Figure 20: Summary of Price 2 variable.

14. Page

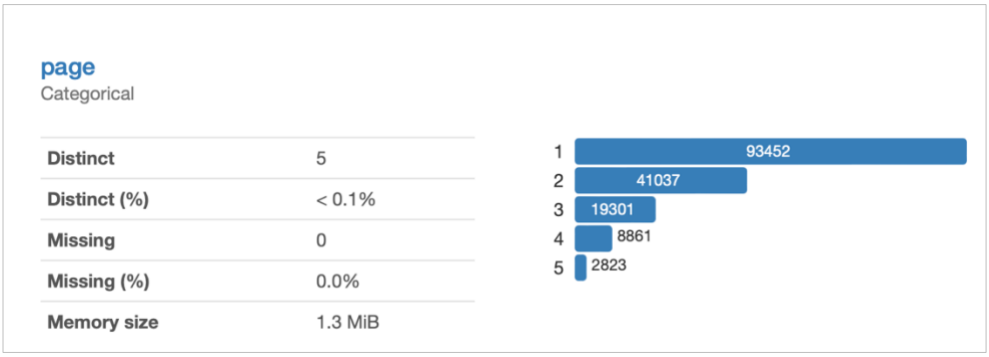


Figure 21: Summary of the Page variable.

Market Basket Analysis

Market basket analysis using association rules were accomplished using the Apriori, Eclat and Weclat algorithms in R. To compare the results provided by each algorithm, support, confidence and minimum length of the itemsets were kept consistent for all the algorithms.

Support is defined how likely the left-hand side clothing models and right-hand side clothing models, indicated by product codes, occur together in a customer session on the e-commerce store. Since, these items are commonly occurring together, pricing strategies such as bundling for these items may be advantageous from a demand generation and sales perspective.

Confidence is defined by when the products on the left-hand side are visited in a customer session, how likely the right-hand side clothing models pages will also be visited during the same customer session.

Lift is defined by the factor by which the probability of visiting the product pages on the right-hand side clothing models increases if the clothing model pages on the left-hand side have already been visited. These association rules can allow for the best clothing models to be recommended to new user sessions on the e-commerce website based on this historic data. With a wide array of clothing models, 217 distinct clothing models to be exact, recommending the best clothing model product page based on a customer's behaviour can be a valuable tactic to cross-sell and increase average order values.

Some ways to implement this information would be to provide additional integrations into the current ecommerce to include product recommendations based on which product pages were visited or send out communications through platforms such as email marketing with products they may be interested in based on previous sessions.

Apriori Algorithm

```
inspect(head(rules_weclat, n = 5, by = "support"))
```

```
##     lhs      rhs   support    confidence lift     itemset
## [1] {A5}  => {A2}  0.03637726 0.4484351  4.213571 3116
## [2] {A2}  => {A5}  0.03637726 0.3418068  4.213571 3116
## [3] {B13} => {B10} 0.03629401 0.4871508  4.846495 3047
## [4] {B10} => {B13} 0.03629401 0.3610766  4.846495 3047
## [5] {A2}  => {A1}  0.03583618 0.3367227  3.845105 3136
```

```
inspect(head(rules_weclat, n = 5, by = "confidence"))
```

```
##     lhs                                rhs  support     confidence lift
## [1] {A10, A2, A3, A6, A7, A8, A9}  => {A5} 0.005410805 0.9629630  11.870779
## [2] {A11, A5, A7, A8, A9}          => {A2} 0.005244319 0.9618321   9.037535
## [3] {A12, A2, A3, A6, A7, A8, A9}  => {A5} 0.005202697 0.9615385  11.853219
## [4] {A1, A5, A6, A7, A8, A9}       => {A3} 0.005077832 0.9606299  14.194400
## [5] {A10, A12, A2, A3, A6, A7, A9} => {A5} 0.005077832 0.9606299  11.842019
##     itemset
## [1] 1482
## [2] 1595
## [3] 1478
## [4] 1485
## [5] 2028
```

```
inspect(head(rules_weclat, n = 5, by = "lift"))
```

```
##     lhs         rhs   support     confidence lift     itemset
## [1] {P3, P7} => {P8}  0.005119454 0.5885167  23.60551 207
## [2] {P55}    => {P60} 0.006492966 0.3959391  21.18671  42
## [3] {P60}    => {P55} 0.006492966 0.3474388  21.18671  42
## [4] {P3, P8} => {P7}  0.005119454 0.4842520  20.62879 207
## [5] {A20}    => {A19} 0.005452427 0.2752101  20.09786   2
```

```
inspect(head(rules_weclat, n = 5, by = "itemset"))
```

```
##     lhs         rhs   support     confidence lift     itemset
## [1] {B4}     => {A2}  0.018521602 0.1385430  1.301773 3170
## [2] {A2}     => {B4}  0.018521602 0.1740321  1.301773 3170
## [3] {B4}     => {A11} 0.019978357 0.1494396  1.463095 3169
## [4] {A11}    => {B4}  0.019978357 0.1955990  1.463095 3169
## [5] {A2, B4} => {A11} 0.006825939 0.3685393  3.608201 3168
```

Figure 22: Top 5 Association Rules sorted by Support, Confidence, Lift & Count of Itemsets for the Apriori Algorithm

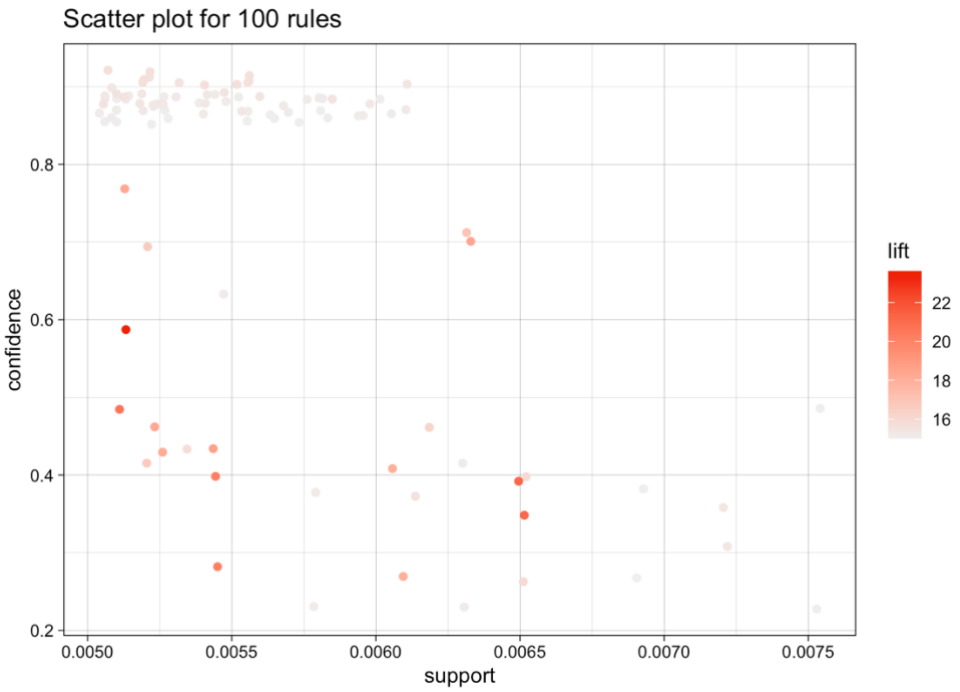


Figure 23: Scatterplot for top 100 rules by for the Apriori Algorithm

Eclat Algorithm

```
inspect(head(rules_eclat, n = 5, by = "support"))
```

```
##     lhs      rhs   support    confidence lift     itemset
## [1] {A5}  => {A2}  0.03637726 0.4484351  4.213571 3112
## [2] {A2}  => {A5}  0.03637726 0.3418068  4.213571 3112
## [3] {B13} => {B10} 0.03629401 0.4871508  4.846495 2712
## [4] {B10} => {B13} 0.03629401 0.3610766  4.846495 2712
## [5] {A2}  => {A1}  0.03583618 0.3367227  3.845105 3145
```

```
inspect(head(rules_eclat, n = 5, by = "confidence"))
```

```
##     lhs                                rhs  support     confidence lift
## [1] {A10, A2, A3, A6, A7, A8, A9}  => {A5} 0.005410805 0.9629630  11.870779
## [2] {A11, A5, A7, A8, A9}          => {A2} 0.005244319 0.9618321   9.037535
## [3] {A12, A2, A3, A6, A7, A8, A9}  => {A5} 0.005202697 0.9615385  11.853219
## [4] {A1, A5, A6, A7, A8, A9}       => {A3} 0.005077832 0.9606299  14.194400
## [5] {A10, A12, A2, A3, A6, A7, A9} => {A5} 0.005077832 0.9606299  11.842019
##     itemset
## [1] 1420
## [2] 1524
## [3] 1409
## [4] 1424
## [5] 1613
```

```
inspect(head(rules_eclat, n = 5, by = "lift"))
```

```
##     lhs         rhs   support     confidence lift     itemset
## [1] {P3, P7} => {P8}  0.005119454 0.5885167  23.60551 333
## [2] {P55}    => {P60} 0.006492966 0.3959391  21.18671  53
## [3] {P60}    => {P55} 0.006492966 0.3474388  21.18671  53
## [4] {P3, P8} => {P7}  0.005119454 0.4842520  20.62879 333
## [5] {A20}    => {A19} 0.005452427 0.2752101  20.09786   1
```

```
inspect(head(rules_eclat, n = 5, by = "itemset"))
```

```
##     lhs     rhs  support    confidence lift     itemset
## [1] {B4} => {A2} 0.01852160 0.1385430  1.301773 3170
## [2] {A2} => {B4} 0.01852160 0.1740321  1.301773 3170
## [3] {P1} => {A2} 0.01914593 0.2030009  1.907430 3169
## [4] {A2} => {P1} 0.01914593 0.1798983  1.907430 3169
## [5] {P1} => {B4} 0.01698160 0.1800530  1.346810 3168
```

Figure 24: Top 5 Association Rules sorted by Support, Confidence, Lift & Count of Itemsets for the Eclat Algorithm

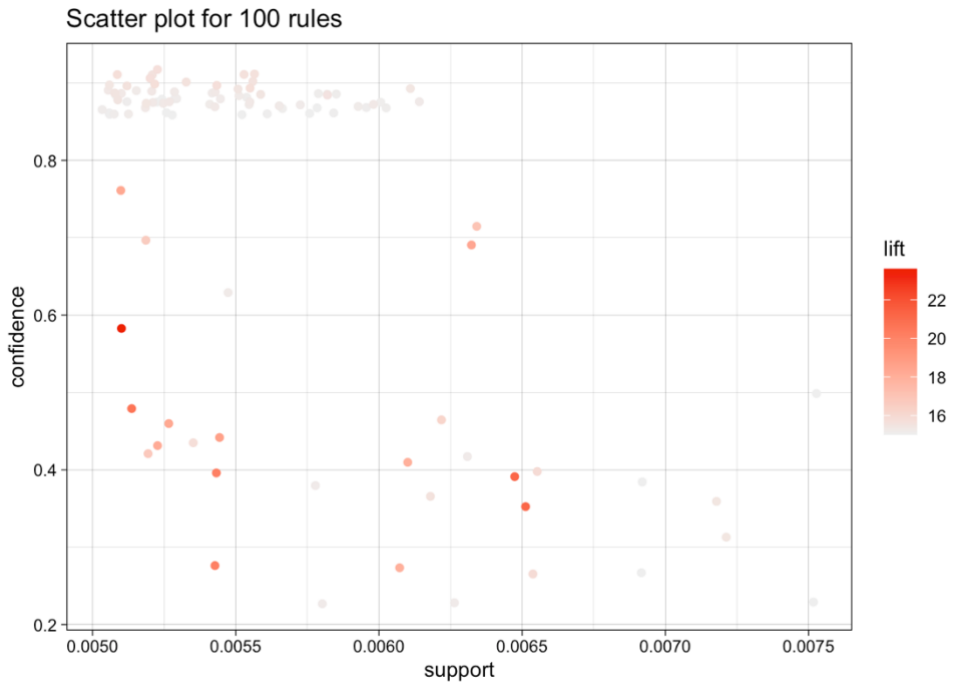


Figure 25: Scatterplot for top 100 rules by for the Eclat Algorithm

Weclat Algorithm

```
inspect(head(rules_weclat, n = 5, by = "support"))
```

```
##     lhs      rhs   support    confidence lift     itemset
## [1] {A5}  => {A2}  0.03637726 0.4484351  4.213571 3116
## [2] {A2}  => {A5}  0.03637726 0.3418068  4.213571 3116
## [3] {B13} => {B10} 0.03629401 0.4871508  4.846495 3047
## [4] {B10} => {B13} 0.03629401 0.3610766  4.846495 3047
## [5] {A2}  => {A1}  0.03583618 0.3367227  3.845105 3136
```

```
inspect(head(rules_weclat, n = 5, by = "confidence"))
```

```
##     lhs                                rhs  support     confidence lift
## [1] {A10, A2, A3, A6, A7, A8, A9}  => {A5} 0.005410805 0.9629630  11.870779
## [2] {A11, A5, A7, A8, A9}          => {A2} 0.005244319 0.9618321   9.037535
## [3] {A12, A2, A3, A6, A7, A8, A9}  => {A5} 0.005202697 0.9615385  11.853219
## [4] {A1, A5, A6, A7, A8, A9}       => {A3} 0.005077832 0.9606299  14.194400
## [5] {A10, A12, A2, A3, A6, A7, A9} => {A5} 0.005077832 0.9606299  11.842019
##     itemset
## [1] 1482
## [2] 1595
## [3] 1478
## [4] 1485
## [5] 2028
```

```
inspect(head(rules_weclat, n = 5, by = "lift"))
```

```
##     lhs         rhs   support     confidence lift     itemset
## [1] {P3, P7} => {P8}  0.005119454 0.5885167  23.60551 207
## [2] {P55}    => {P60} 0.006492966 0.3959391  21.18671  42
## [3] {P60}    => {P55} 0.006492966 0.3474388  21.18671  42
## [4] {P3, P8} => {P7}  0.005119454 0.4842520  20.62879 207
## [5] {A20}    => {A19} 0.005452427 0.2752101  20.09786   2
```

```
inspect(head(rules_weclat, n = 5, by = "itemset"))
```

```
##     lhs         rhs   support     confidence lift     itemset
## [1] {B4}     => {A2}  0.018521602 0.1385430  1.301773 3170
## [2] {A2}     => {B4}  0.018521602 0.1740321  1.301773 3170
## [3] {B4}     => {A11} 0.019978357 0.1494396  1.463095 3169
## [4] {A11}    => {B4}  0.019978357 0.1955990  1.463095 3169
## [5] {A2, B4} => {A11} 0.006825939 0.3685393  3.608201 3168
```

Figure 26: Top 5 Association Rules sorted by Support, Confidence, Lift & Count of Itemsets for the Weclat Algorithm

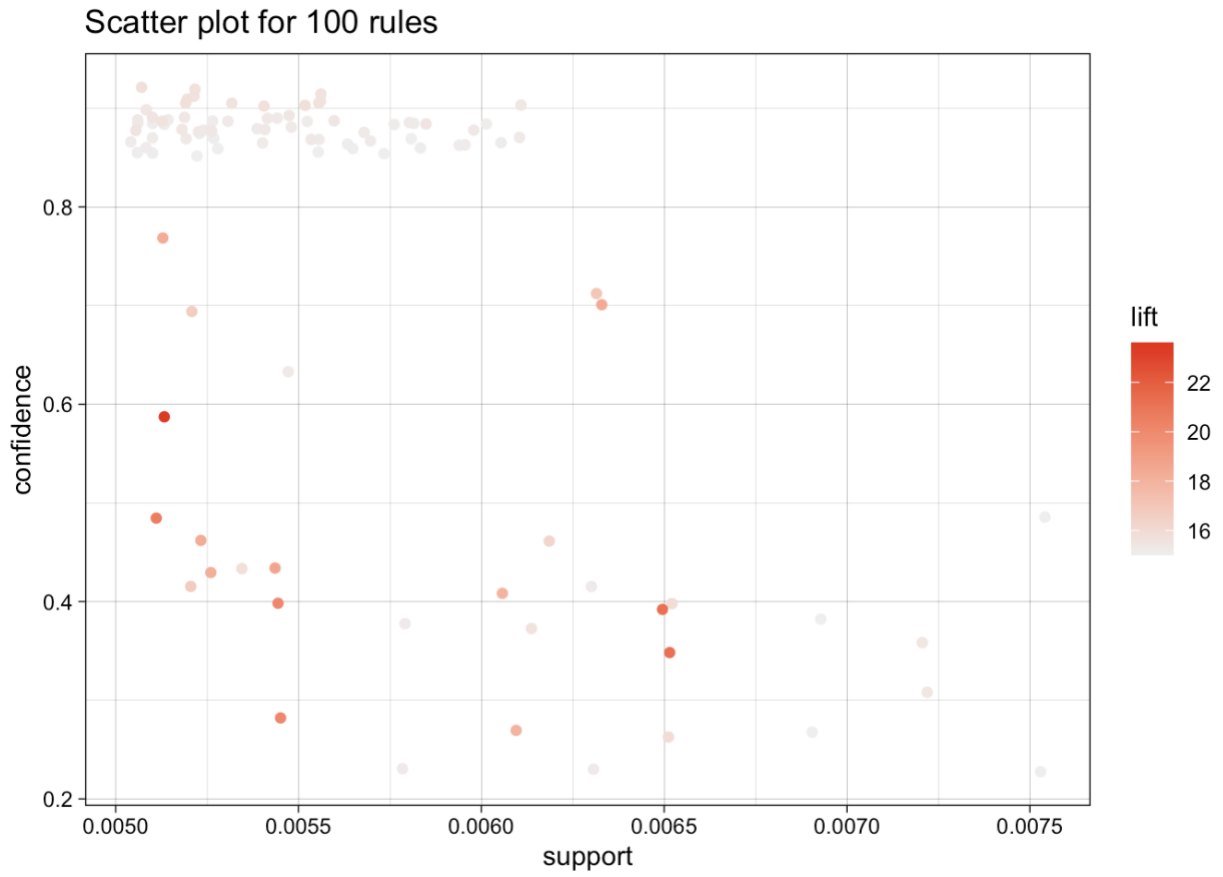


Figure 27: Scatterplot for top 100 rules by for the Weclat Algorithm

These association rules, provide important insight into what products were viewed together. As see in figures 22 – 27, the products viewed together most can be found by the association rules with the highest support.

From the association rules retrieved from running these algorithms, the best product to propose to the customer is recommending the highest confidence right-hand side items to the items which they have already viewed which can be interpreted as the left-hand side items. In the case where their itemset is not in the association rules with the given support and confidence thresholds, the next closest candidate itemset with the highest confidence should be used to recommend products.

When comparing the Apriori, Eclat and Weclat for market basket analysis they provide the same or similar results; it is very evident when comparing the scatterplots for each that were created using the top 100 association rules by lift Although their results are equivalent their execution time and memory usage may differ.

The main difference between these three algorithms is the way they search and how they handle reoccurring. Apriori uses a broad search strategy and achieves candidate itemsets with larger sizes and prunes the infrequent ones based on a minimum support threshold. Eclat uses a depth-first search strategy and combines itemsets that contain a particular item. Weclat is an extension of Eclat that considers the weight of the items in the dataset.

Classification Analysis

Feature selection was performed using HPSPLIT analysis in SAS. The selected features are: Page 2 (clothing model), and Colour, for the predicted variable of Price 2.

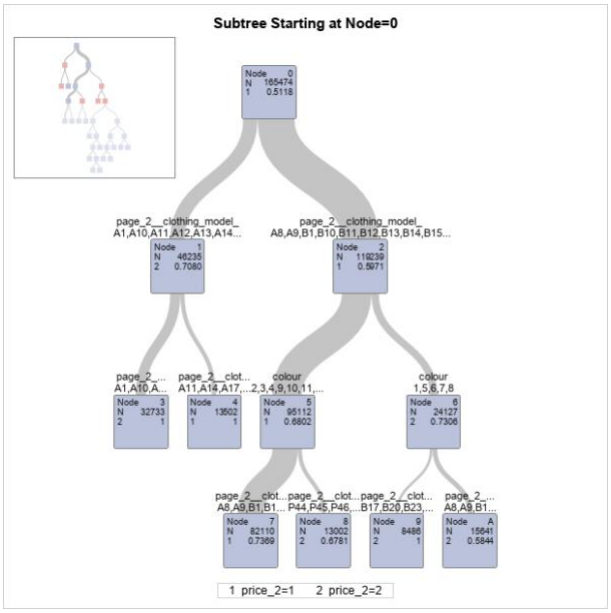


Figure 28: Classifier decision trees made using HPSplit in SAS of clickstream data

The HPSPLIT Procedure

Model-Based Confusion Matrix

Actual	Predicted 1	Predicted 2	Error Rate
1	84695	0	0.0000
2	0	80779	0.0000

Model-Based Fit Statistics for Selected Tree

N Leaves	ASE	Mis-class	Sensitivity	Specificity	Entropy	Gini	RSS	AUC
17	0	0.0000	1.0000	1.0000	0	0	0	1.0000

Figure 28: Confusion matrix for classifier decision trees made using HPSplit in SAS of clickstream data

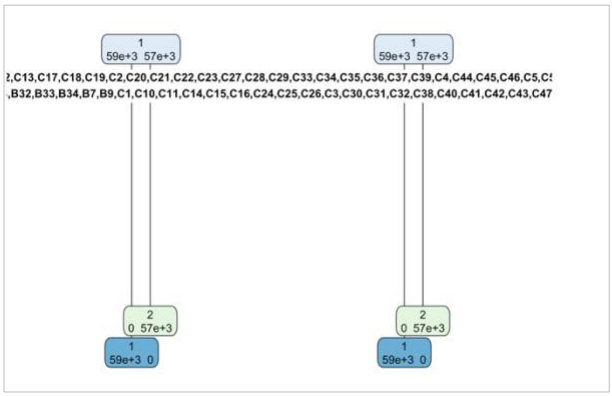


Figure 29: Classifier decision trees made using Rpart in R of clickstream data; first model (left) uses all features and second model (right) uses all features selected from HPsplit

```
Confusion Matrix and Statistics

          Reference
Prediction     1     2
         1 33761 31627
         2 40669 42870

               Accuracy : 0.5146
                 95% CI : (0.512, 0.5171)
    No Information Rate : 0.5002
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.0291

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.4536
            Specificity : 0.5755
         Pos Pred Value : 0.5163
         Neg Pred Value : 0.5132
             Prevalence : 0.4998
         Detection Rate : 0.2267
   Detection Prevalence : 0.4391
      Balanced Accuracy : 0.5145

       'Positive' Class : 1
```

```
Confusion Matrix and Statistics

          Reference
Prediction     1     2
         1 33761 31627
         2 40669 42870

               Accuracy : 0.5146
                 95% CI : (0.512, 0.5171)
    No Information Rate : 0.5002
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.0291

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.4536
            Specificity : 0.5755
         Pos Pred Value : 0.5163
         Neg Pred Value : 0.5132
             Prevalence : 0.4998
         Detection Rate : 0.2267
   Detection Prevalence : 0.4391
      Balanced Accuracy : 0.5145

       'Positive' Class : 1
```

Figure 30: Confusion matrix for classifier decision trees made using Rpart in R of clickstream data; first model (left) uses all features and second model (right) uses all features selected from HPsplit

As per image above, future prices of products can be predicted based on the classification decision trees which 2 important features which are primarily indicative of price: the types of clothing models (Page 2 Clothing Model feature) and the colour of the clothing models (colour feature). Therefore, for future new clothing models that are going to be sold at the e-commerce store, assuming the clothing model exists and may be produced in a new colour, the price can be predicted using this model. Also, these results were obtained using 10-fold cross validation to help overfitting and find a balance between bias and variance.

Lastly, when comparing the confusion matrix, the HPsplit decision trees perform better with higher accuracy

A limitation for this classification technique is the prediction of price being dependent on characteristics such as clothing model may affect long term results as commonly clothing companies follow the latest fashion trends and often release new items seasonally and slowly eliminate last season's clothing models. In this case, classification to determine if price is below or above average will only be accurate for the specific time frame the items are all sold, not including companies that may have staple items that are sold year-round, year over year. Again, these results might not be as accurate when adjusting for long term economical changes such as inflation.

Cluster Analysis

Cluster analysis was done using the Kmode clustering algorithm in R and Kmeans clustering algorithm in python.

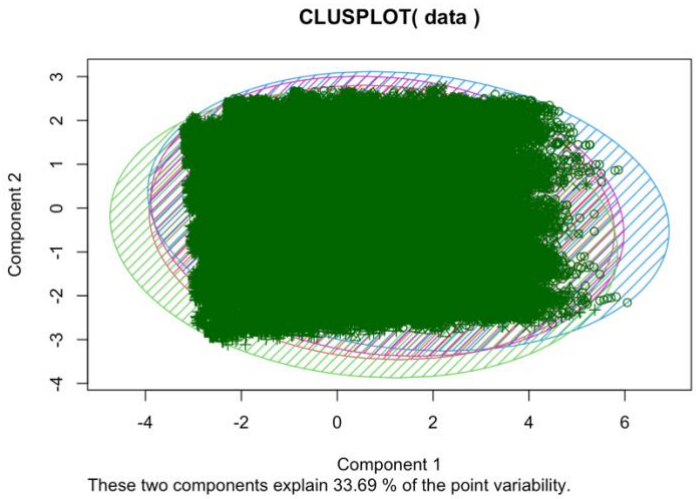


Figure 31: Cluster of clickstream data using Kmode Clustering Algorithm in R

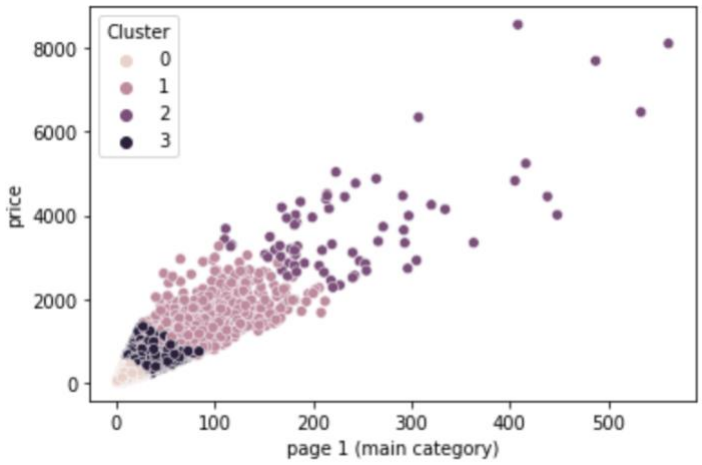


Figure 32: Cluster of clickstream data using Kmeans Clustering Algorithm in Python

As per the figures above, a clear differentiation can be visualized by the plot created using the Kmeans. In this plot, 4 distinct clusters can be visualized and be used to create customer segments based on similar customer session behaviour, specifically price and categories of product viewed during sessions.

In the Kmode cluster plot, unlike the Kmeans cluster plot, the clusters are not as easy to visualize and there is overlapping but the clusters are differentiated by the colored ovals. This clustering algorithm

minimizes the sum of the dissimilarities between the data points and their respective centroids in contrast. In conclusion, Kmode and Kmeans clustering algorithms can both be used to differentiate the datasets into customer segments using different techniques.

Ethical Considerations

In this analysis, privacy rights, data validity, and algorithm fairness are important factors to consider when dealing with user driven data which monitor user interactions. In this case, there was no Personally Identifiable Information (PII) and did not violate anyone's privacy rights and generally is compliant with General Data Protection Regulation (GDPR). However, in this case because of the nature of when this data was recorded, and the growing concern on user permissions when collecting data, there is no way to guarantee if this data was collected with user permission despite the date being anonymized. Data validity is assumed as all data points were complete with no missing values including no Null and NaN values and assumes this is an accurate distribution of data points. Lastly, the algorithms used in this analysis are isolated to the dataset and are not introducing external bias that some deep learning models may have, but there are still areas for bias including confirmation bias, availability bias and stability bias that can be introduced preliminarily and after the machine learning result computations.

Limitations & Recommendations

This dataset is limited by how many different features are collected. Clickstream datasets can collect far more features and attributes and can also potentially be used with other datasets collected to and originate to customer specific information. For example, having customer information like how many times a product page was view, number of orders placed, amount spent, wider time ranges to see peaks in traffic and customer activity and all shed more information on the overall customer journey on the e-commerce website.

Lastly, another limitation is not being knowing more specific details about the design of the different clothing models, whether as an image or by description that provides more information on the style. In the original study, product images were analyzed when conducting marketing basket analysis.

As mentioned previously, it would be interesting to gather more information such as what the styles of the clothing models looked like as well as gathering more information from this clickstream data. Having a more comprehensive dataset with more features regarding product styles, as well as more information of customer behaviour such as duration of sessions, clickthrough and buy-through rates, as well as past order values all can help provide more insight into the customer journey on this e-commerce website.

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Appendix

Dataset: first 6 rows

	year	month	day	order	country	sessionID	page1	page2	colour	location	modelphotography	price	price2	page
1	2008	4	1	1	29	1	1	A13	1	5	1	28	2	1
2	2008	4	1	2	29	1	1	A16	1	6	1	33	2	1
3	2008	4	1	3	29	1	2	B4	10	2	1	52	1	1
4	2008	4	1	4	29	1	2	B17	6	6	2	38	2	1
5	2008	4	1	5	29	1	2	B8	4	3	2	52	1	1
6	2008	4	1	6	29	1	3	C56	6	1	2	57	1	4

Countries

Variable indicating the country of origin of the IP address with the following categories:

1-Australia

2-Austria

3-Belgium

4-British Virgin Islands

5-Cayman Islands

6-Christmas Island

7-Croatia

8-Cyprus

9-Czech Republic

10-Denmark

11-Estonia

12-unidentified

13-Faroe Islands

14-Finland

25-Luxembourg

26-Mexico

27-Netherlands

28-Norway

29-Poland

30-Portugal

31-Romania

32-Russia

33-San Marino

34-Slovakia

35-Slovenia

36-Spain

37-Sweden

38-Switzerland

15-France

16-Germany

17-Greece

18-Hungary

19-Iceland

20-India

21-Ireland

22-Italy

23-Latvia

24-Lithuania

39-Ukraine

40-United Arab Emirates

41-United Kingdom

42-USA

43-biz (*.biz)

44-com (*.com)

45-int (*.int)

46-net (*.net)

47-org (*.org)

Colours

1-beige

2-black

3-blue

4-brown

5-burgundy

6-gray

7-green

8-navy blue

9-of many colors

10-olive

11-pink

12-red

13-violet

14-white